SHAGITHTHIAH KIRUPAKARAN

shagikirupa16@gmail.com | +94 74 360 5457 | LinkedIn | GitHub

Education

- **University of Peradeniya** *Jun 2023 - Present*
  BSc. Eng. (Hons.) in Computer Engineering, Current GPA: 3.6/4.00 Kandy, Sri Lanka

Experience

- **Crisis Lab** *Apr 2026 - Present*
  Research Intern - Remote Wellington, New Zealand
  - **Tech Stack:** Python, PyTorch, ResNet18, Label Studio, Groq API
  - Built a first-of-its-kind annotated ground-level flood severity image dataset for Sri Lanka, sourcing ∼7,500 records from the NIWA Citizen Science API and using LLM-based classification for damage relevance and severity labeling.
  - Trained a ResNet18 transfer learning model for 4-class flood severity classification, achieving ∼73% validation accuracy, with checkpointing and learning-rate scheduling.
  - Designing a 5-layer citizen science app architecture (mobile app, API gateway, ML classifier, geo-tagged database, responder dashboard) integrating AI-based severity classification into the reporting pipeline, a capability absent in comparable international flood-reporting apps.
  - Co-designing the annotation workflow and research direction with remote mentor through weekly check-ins, working toward a paper submission alongside the dataset and model contributions.

- **Undergraduate Teaching Assistant** *Dec 2024 - Mar 2025*
  CO1010 Programming for Engineers Department of Computer Engineering, UoP
  - Assisted undergraduates in mastering Python, covering topics from fundamental concepts to object-oriented programming (OOP).
  - Provided hands-on guidance to make learning engaging and practical, helping students build coding confidence and improve their problem-solving skills.

Selected Projects

- **Cross-Resource Interference in Heterogeneous ARM Systems** *Sep 2026 - Present*
  Linux, Python, C/C++, Bash, perf, Hardware Performance Counters (PMU), NVIDIA Profiling Tools Final Year Project
  - Investigating **CPU-GPU-memory interference in heterogeneous systems**, focusing on performance degradation caused by contention for shared resources such as DRAM bandwidth, cache, memory hierarchy, interconnects, power, and thermal capacity.
  - Designing controlled performance experiments comparing isolated and co-running CPU, GPU, and memory-intensive workloads, using execution time, IPC, cache misses, memory bandwidth, CPU stalls, and GPU utilization to identify the causes of cross-resource interference.
  - Applying Linux performance-engineering techniques including hardware performance counters, CPU affinity, workload co-execution, resource controls, and profiling to characterize and investigate methods for reducing interference.
  - Working toward validation of the interference analysis on heterogeneous ARM/Linux platforms using realistic **Nanopore genomics workloads**, including GPU-intensive basecalling and CPU/memory-intensive downstream genomic processing.

- **SmartIV - Intelligent IV Drip Monitoring System (Group Project)** *Jan 2026 - Present*
  React Native, AWS IoT Core, MQTT, Tauri (Rust) 3rd Year Project
  - Engineered the complete mobile application front-end using React Native, delivering an intuitive interface for real-time patient monitoring, automated alerts, and historical session tracking.
  - Architected and managed all cloud infrastructure and hosting, configuring AWS IoT Core and MQTT for seamless, real-time data streaming from the hardware edge devices.
  - Implemented secure user authentication via AWS Cognito and ensured the reliable deployment and maintenance of the system's backend services.

- **Voting Dynamics Using Agent-Based Modelling (Personal Project)**
  Python, NumPy, NetworkX, Matplotlib Dec 2025
  - Built an Agent-Based Model simulating how opinions spread and polarize across social networks.
  - Modeled voter behavior using social influence, noise, and zealotry as key parameters.
  - Found that even a small group of zealots can block consensus, mirroring real-world social-media polarization.
- **EngEx 2025 Registration Portal – Smart Exhibition Management System (Group Project)** *Aug 2025 - Nov 2025*
  React, Node.js, Express, PostgreSQL 2nd Year Project
  - Built an RFID-based visitor registration portal for EngEx 2025, the university's annual engineering exhibition.
  - Developed the Authentication & Registration module: secure login, JWT sessions, and individual/batch visitor registration.
  - Integrated the module with the RFID tag-linking system for real-time use during the live exhibition.

Achievements & Awards

- **Winner - SLASSCOM National Ingenuity Awards 2026** *Jun 2026*
  Best Innovative Product (University Category) – Central Province
  - Awarded for Smart IV, an IoT-based system designed to monitor IV drip flow in real-time, estimate remaining volume, and instantly alert healthcare staff.
- **1st Runner-Up - NetX IoT Challenge** *Sep 2026*
  Team Zephyrus – IEEE Student Branch, University of Sri Jayewardenepura
  - Secured 1st Runner-Up at the NetX IoT Challenge, demonstrating innovation, technical implementation, and teamwork in developing an IoT-based solution.
- **Finalist - SLIoT Challenge** *April 2026*
  Smart IV Project
  - Demonstrated the hardware and software prototype to industry experts at the national finals, showcasing its real-time IoT capabilities.
- **Finalist - Startup Spark, TECHNO 2024** *Oct 2024*
  Intelligent Traffic Monitoring System (ITMS)
  - Pitched an AI-driven traffic management solution featuring dynamic signal control, emergency vehicle prioritization, and real-time accident detection to industry experts and potential investors.
- **University Colours - Chess** *2024*
  University of Peradeniya
  - Awarded prestigious University Colours in recognition of outstanding performance as a key member of the Women's Chess Team that secured the Championship title at the Inter-University Games 2024.

Extra Curricular Activities

- **National Arbiter** *2025- Present*
  Chess
  - Passed the National Arbiter certification exam in 2022 and officially awarded the title in 2025 after successfully fulfilling all required tournament norms.
- **Team Player** *Jun 2024 - Apr 2025*
  Women's Chess Team, University of Peradeniya
  - Secured 1st Place at the Inter-University Championship 2024, representing the university and collaborating on team strategies and match preparations.
- **Captain** *2024 - Present*
  E21 Batch Women's Chess Team
  - Led the batch team to a Runners-Up finish at the Freshers Meet 2024, organizing practice sessions and fostering a collaborative environment for intra-faculty championships.

- **Event Host (Compering) & Singer** *2023 - Present*
  Various University Events
  - Hosted and performed at numerous university functions, demonstrating strong public speaking, crowd engagement, and stage presence.
- **Secretary** *2021*
  Advanced Level Student's Union, Union College Tellippalai
  - Managed administrative duties, coordinated student-led initiatives, and served as the main point of contact between students and school administration.
  - Organized and successfully executed the farewell ceremony for the 2021 Advanced Level graduating batch.

Volunteer Experience

- **Research Volunteer** *May 2025 - Jul 2025*
  Volunteer Multidisciplinary AI Research Centre (MARC)
  - Contributed to a machine learning-based slum detection project, utilizing high-resolution satellite imagery to identify and map informal settlements.
  - Annotated and verified spatial data using Google Earth Pro, cross-referencing satellite and street-level views to construct a reliable ground-truth dataset for model training.
- **Media Team Member** *Nov 2025 - Present*
  Team Member Artificial Intelligence Forum for Students (AIFS)
  - Assisted in student engagement for university AI events and workshops.
- **Student Volunteer** *Nov 2025 - Present*
  Volunteer PeraCom Seminar Organizing Team
  - Assisted in guest coordination, and Students arrangement for department-wide computing and engineering seminars.